add the numbers if we are going east, and subtract if we are going west. Here are some examples of how we may need to calculate the difference of time zones.

If you are in London at 12:00, and want to know what time it is in Japan, you would need to first figure out that London is 0 degrees (right on the prime meridian), and Japan is 135 degrees East. So the difference is 135 degrees (135–0), divided by 15 which equals 9. It means there is a 9-hour difference between London and Japan. Since Japan is further east than London is, you would add 9 hours to 12:00. The answer is at 12:00 noon London time, it is 9:00pm in Japan.

Now we suppose imagine that we are going through the International Date Line. Pretend you are in Japan, which is 135 degrees east and you wanted to know what time it is in Hawaii, which is 150 West. Well, there is 45 (180–135) degrees difference between Japan and the IDL. Also there is 30 (180–150) degrees difference between the IDL and Hawaii. Therefore the difference in time is (45 + 30/15 = 5) 5 hours. Now the tricky part is that Japan and Hawaii are on different days. It is one day ahead on the left side of the IDL compared to the right side. If it is 3:00pm in Japan on Thursday that means it is 3:00 + 5 hours = 8:00pm in Hawaii. However notice that when crossing the IDL we subtract a day going east. So, in Hawaii it is 8:00pm on Wednesday.

Now note that Latitudinal lines are imaginary horizontal lines over the Earth's globe. 0° longitudinal line is Equator. Earth completes one rotation on its axis in 24 hours and in the process turns a complete circle of 360°. This means Earth rotates 360°/24 = 15° in one hour. Every gain or loss of 1° longitude stands for 4 minutes.

360° = 24 hours = 1440 min (24 × 60)

Difference of time for 15° longitude = one hour.

Difference of time for 1° longitude = 4 minutes.

**Longitude Calculations Procedures**

a. First locate the two places involved
b. Find the longitude difference
c. Convert the longitude difference to time and,
d. Adjust the time according to the direction of movement, (west or east).

**Example 1**

Ponni starts her journey at longitude 0° at 12 noon and she's moving towards eastward of longitude 10°. Calculate the time that Ponni will arrive at her destination.

***Solution***

Initial time = 12 noon

Destination = 10°E

Conversion of degree to time

1 hour = 15°

and 4 minutes = 1°

Hence 10° = (4 × 10) minutes

= 40 minutes

Destination time = Initial time + calculated time

= 12 noon + 40minutes

= 12:40pm

**Example 2**

If the time at village A (long 75°W) is 5:00 pm on Friday. Calculate the time and day at village B (long 120°E)



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